



ERP Systems

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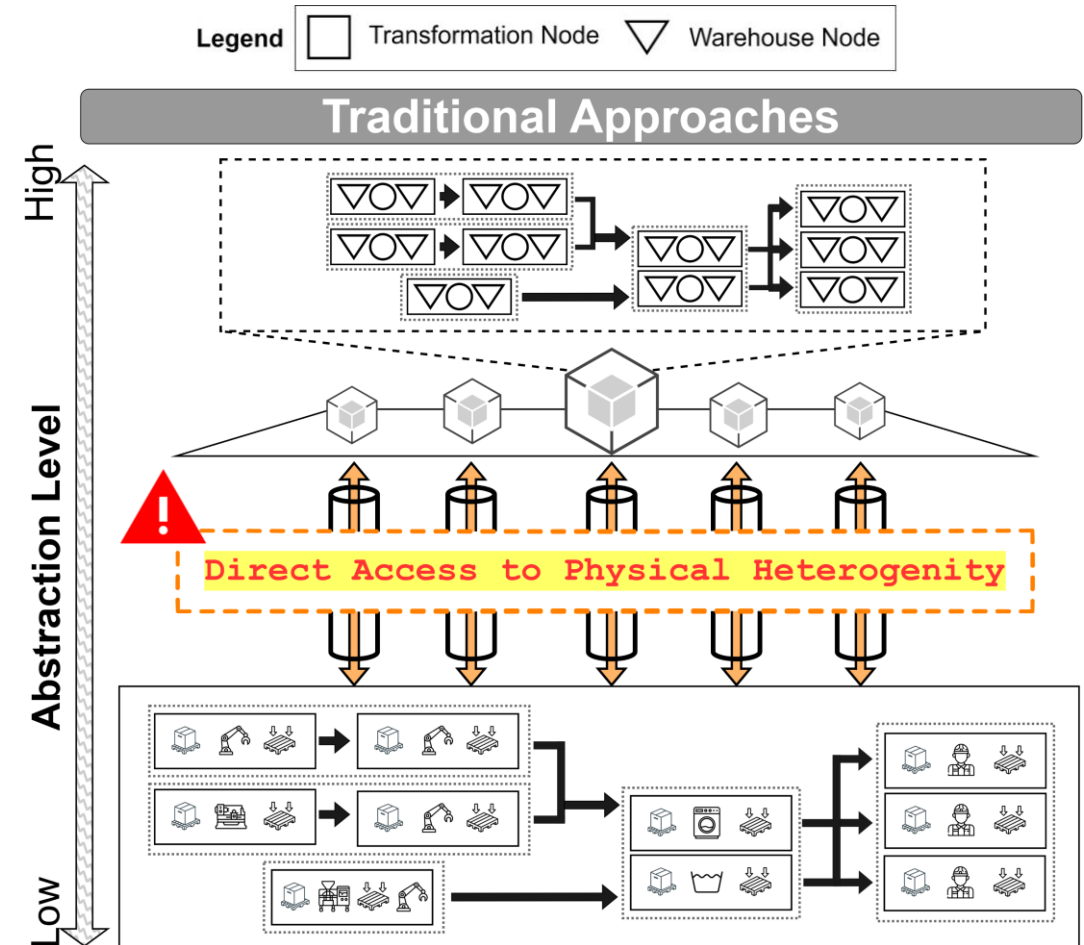
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ERPs: why?

Historically, companies used as information systems the so-called “**islands of automations**”, that is a wide collection of various systems for the management of different business processes ...

- Many possible levels of integration among systems
- Many possible levels of scalability and maintenance

This process of maintaining a multitude of different systems has **blown up** throughout the years, reaching a point where the cost of **updating** and **maintaining** these systems has become **greater** than the cost of implementing and installing a **new** system.



ERPs Challenges



«The notion that a company can and ought to have an expert (or a group of experts) create for it a single, completely integrated supersystem – a Management Information System (MIS) – to help it govern every aspect of its activity is absurd.»

John Dearden, 1972

ERPs Challenges



Moving towards an ERP can be a **complex** process

- Many companies **fail** or do not realize the financial **returns** they had expected for the ERP investment
- This is very **typical** of any new transformation induced by a new (information) technology
- This process can be **long** and can imply **major changes** across different dimensions of the organizational processes of a company

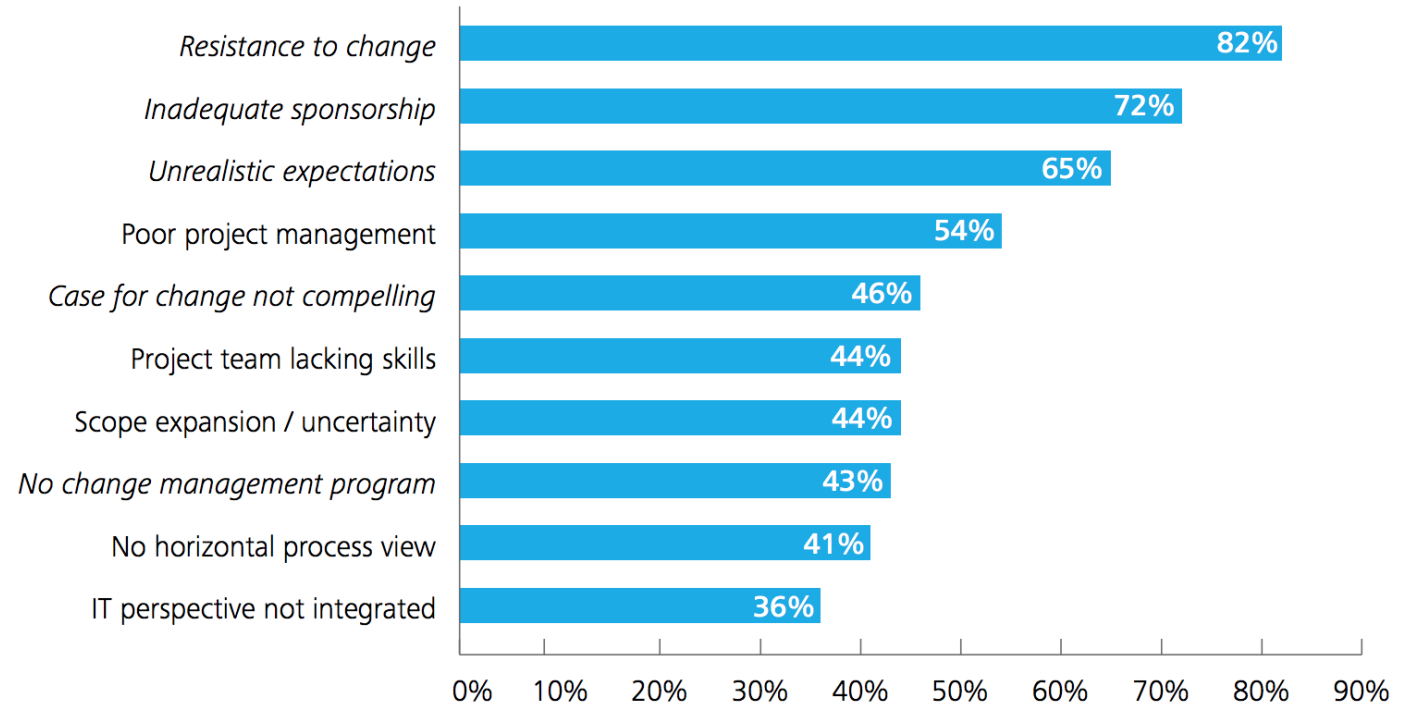
ERPs Challenges

«The single biggest failure point for ERP implementations is the need for change management.»

Neville Turbet, Project Perfect

Gartner estimates that 55% to 75% of **all** ERP projects **fail** to meet their objectives ...

Top 10 barriers to implementation



Source: Deloitte Touche Tohmatsu CIO Survey

MRP I vs. MRP II



ERP systems were born extending the functionalities and the capabilities of

- **Material Requirements Planning (MRP)**, exploiting software solutions for the scheduling of production processes
- **Manufacturing Resource Planning (MRP II)**, consisting in software applications for the coordination of manufacturing processes

An ERP system can be thought as a **suite** of **integrated** applications capable of **collecting**, **storing**, **managing** and **analyzing the data** of a company:

- Continuous update
- Data integration
- Safety and security issues
- Usability and user-friendliness

ERP I in Industry



Under an industrial perspective ...

- ERP I: integrates main ***internal resources*** of an enterprise
 - ❑ **Finance**: accounting, balance sheets, financial resources management ...
 - ❑ **HR**: people administration, payrolls ...
 - ❑ **Production**: planning, stock control ...
 - ❑ **Logistic**: orders management, shipping, supplies ...

Beyond ERP: the ERP II

Initially, ERP systems were designed starting from **back-office** functionalities (no contact with end-users and customers) and only later on they started integrating also **front office** functionalities.

After 2000, the so-called **ERP II** systems were created, with a significant integration into **web-based** applications and **real-time** access to data.

Beyond ERP: the ERP II

Under an industrial perspective ...

- ERP II: evolves from the ERP I, focusing ***also on external collaboration*** and ***full integration*** of the enterprise ecosystem.
 - ❑ **Extended collaboration:** includes relationships with customers, suppliers, partners, and other stakeholders
 - ❑ **Business Intelligence:** integration of advanced data analysis tools to support decision-making based on internal and external data
 - ❑ **Multi-organizational support:** can integrate its functionalities on top of different complex organizational structures, including divisions, business units, different countries and regions
 - ❑ **Integration of new functionalities:** can integrate Customer Relationships solutions (CRM), Supply Chain Management solutions (SCM), e-commerce, and so on
 - ❑ **Web-based**

ERP Modules



ERP systems are typically divided in **modules**, each dedicated to a specific **activity** or **business line**

The main **families of modules** can be identified as

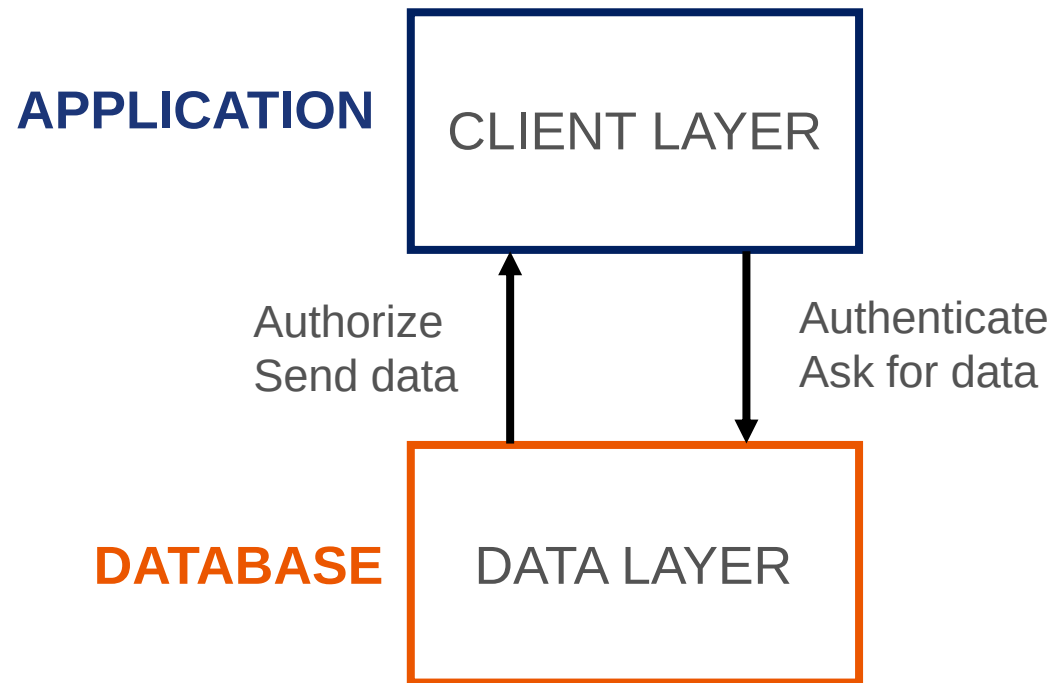
- Finance
- Logistics
- Production & Supply Chain
- Quality Control
- Human Resources
- Sales & Distribution, CRM
- Business Intelligence

Typical **components** of an ERP system include

- Accounting and Bookkeeping
- Planning and Control
- Personnel Management
- Buy/Sell Management
- Stock and Productions
- Supply Chain Management
- Project Management

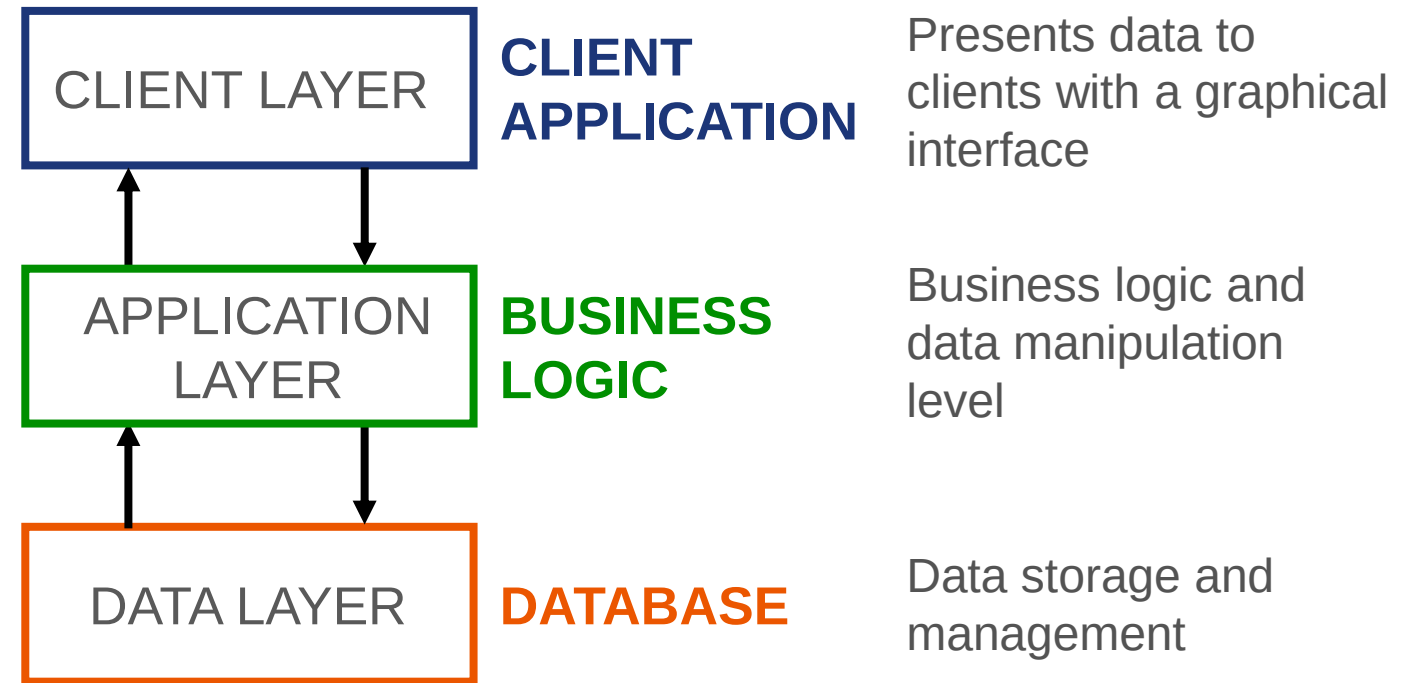
ERP Architecture

Three-Tier Architecture



Three-Tier Architecture

The three layers may run on the same or different machines

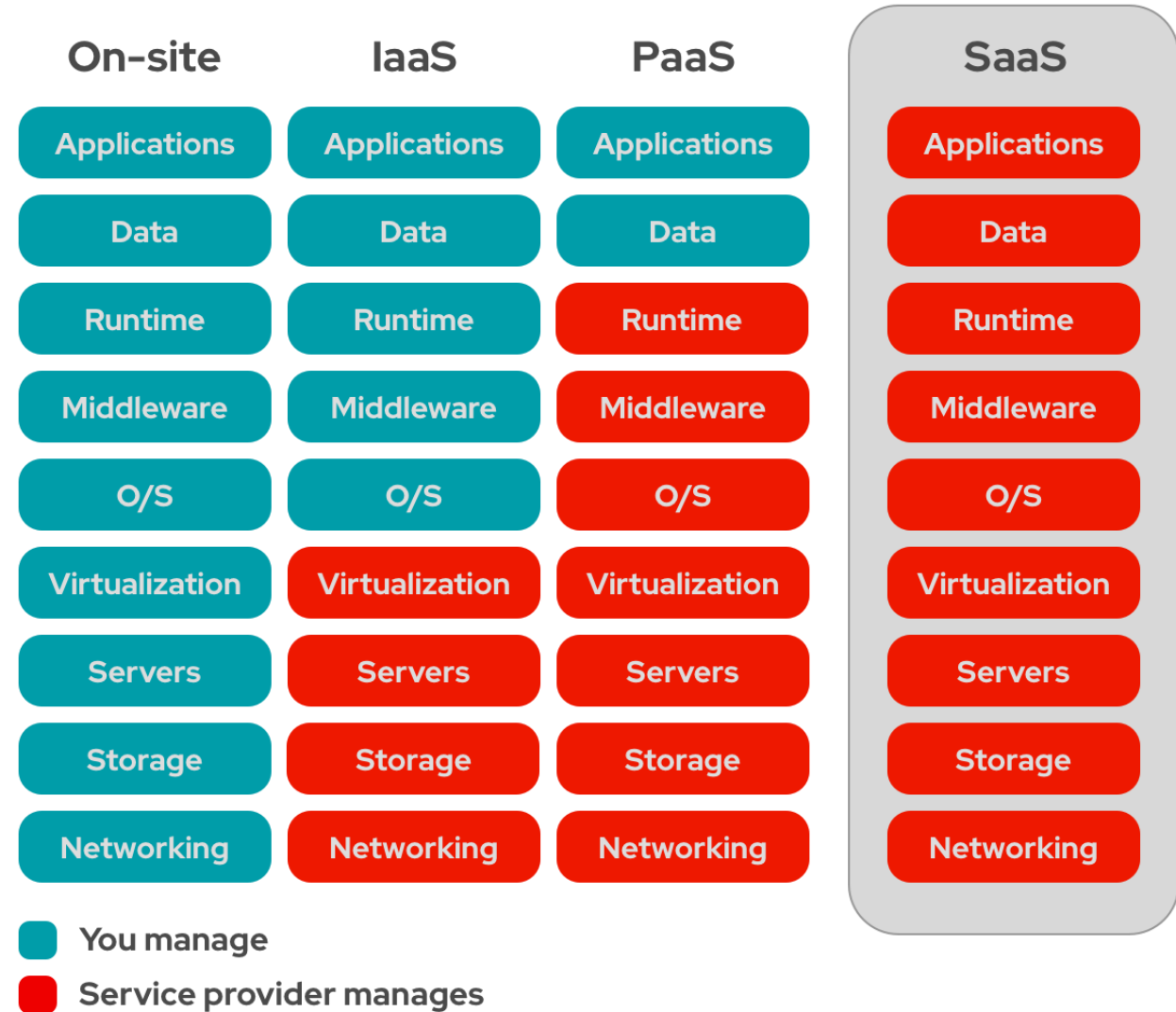


ERP Architecture

Nowadays, ERP is often delivered with the paradigm **Software as a Service (SaaS)**

- **On-demand** software accessed **via web**
- Users buy **only** what they use
- No need for **server installation**
- Model based on **cloud computing**

- (**IaaS**: Infrastructure as a Service)
- (**Paas**: Platform as a Service)



[Source: RedHat]

Advantages



Characteristics assessing the quality of ERP systems:

- **Flexibility** to adapt to enterprise needs
- **Modularity** to scale up with novel business lines
- **Connectivity** to interact to other organizations
- **Continuous update** to the best business practices

Benefits of the use of ERP systems include:

- **Centralization** of information
- **Standardization** of formats
- Transparent information **interchange**
- Personnel **training** on a single application
- Reduction of **lifecycles** and **processing time**
- Reduction of **costs**
- Reduction of **paper documents**

Advantages



Centralization of information:

- **Easy access** to data
- No **multiple** independent data copies and versions
- No **redundant operations** on the same data
- No need to **integrate** multiple data sources to obtain a certain piece of (aggregated) information
- Top-level administrators/managers have a **global view** of what is going on in the company

Standardization of formats:

- Standard **languages**
- Standard **currencies**
- Standard **communication protocols**
- Standard **security levels**

Disadvantages

Drawbacks of the use of ERP systems include:

- Typically, **high costs** for best products
- Too long **return of investment**
- Some level of **customization** is necessary
- **Dependency** on the ERP **producer**
- **Versioning** and update processes
- Excessive and unnecessary **complexity**

Customization



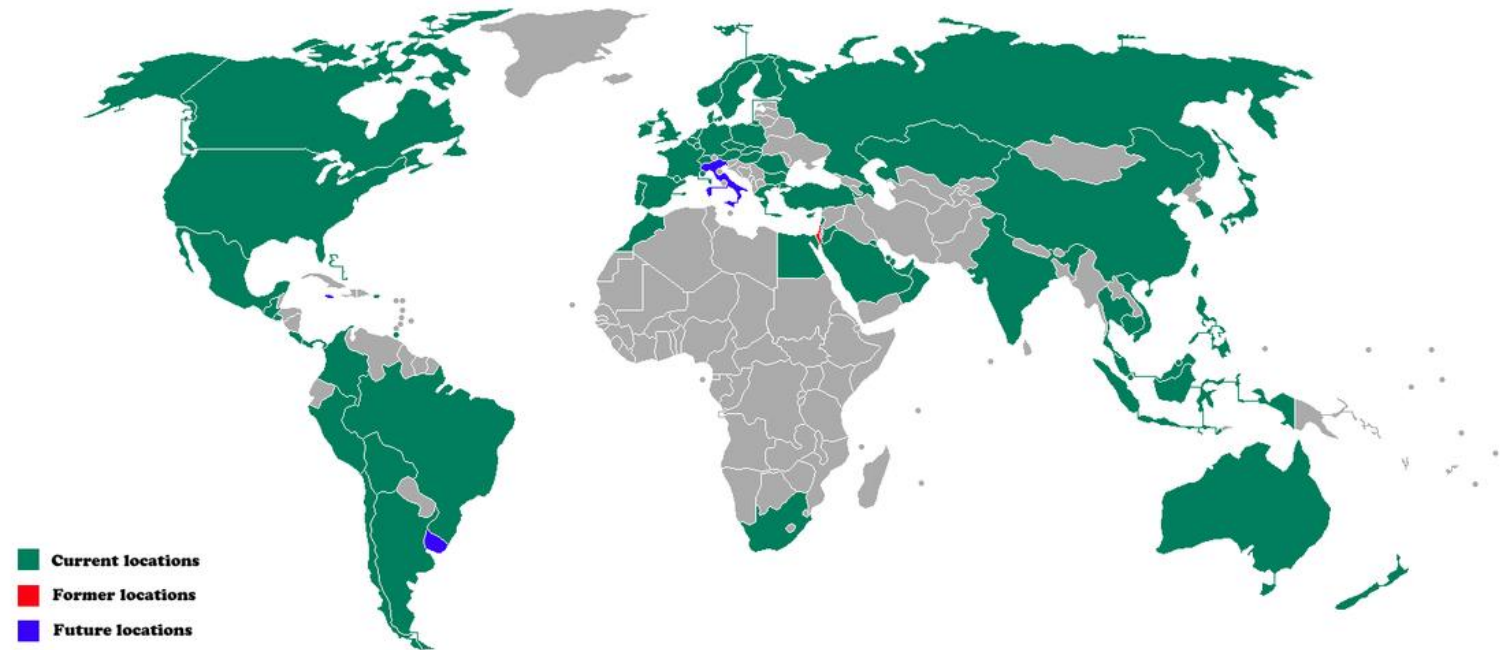
Adapt to the needs of the client:

- Initial **configuration** and **setup**
- Implement and integrate **needed functionalities** (almost always there are some that are **missing...**)
- Re-engineering **business processes**
- **Personnel training**

An Example

Starbucks:

- Global **coffee company** from 1971
- **Worldwide** distribution channels and outlets
- Very complex **supply chain**
- Present in 67 countries



An Example

Main issues with supply chain

- **Over half** of deliveries **late**
- High **outsourcing** costs

After installing ERP system

- **Fixed** issues across **all** supply chain operations
- 500M\$ **savings** in 2 years after implementation
- **Reduced** paperwork, better **forecasting**, ...



ERPs Market Quotas

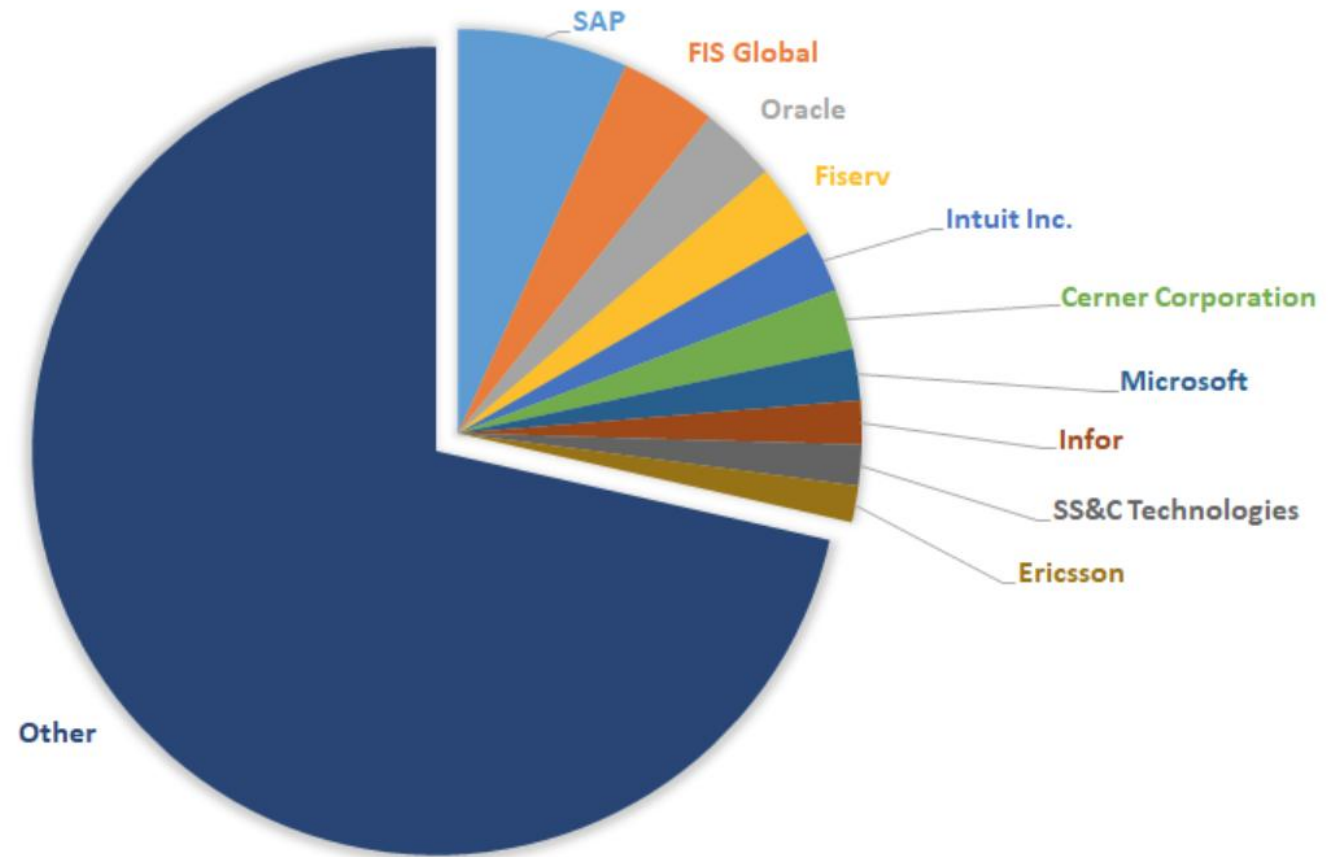
Main **commercial** products:

- SAP ERP
- Zucchetti
- Oracle
- Microsoft Dynamics
- Sage
- ...

Main **open-source** products:

- ERPNext
- Openbravo
- OpenERP
- ADempiere
- iDempiere
- ...

EXHIBIT 1: 2016 ERP APPLICATIONS MARKET SHARES
SPLIT BY TOP 10 ERP VENDORS AND OTHERS, %



ERP system developed by German company SAP

- Latest **version** in 2006 (SAP ERP 6.0)
- Latest **enhancement package** in 2016 (EHP8)

Proprietary and dedicated programming language

- Advanced Business Application Programming (ABAP/4)

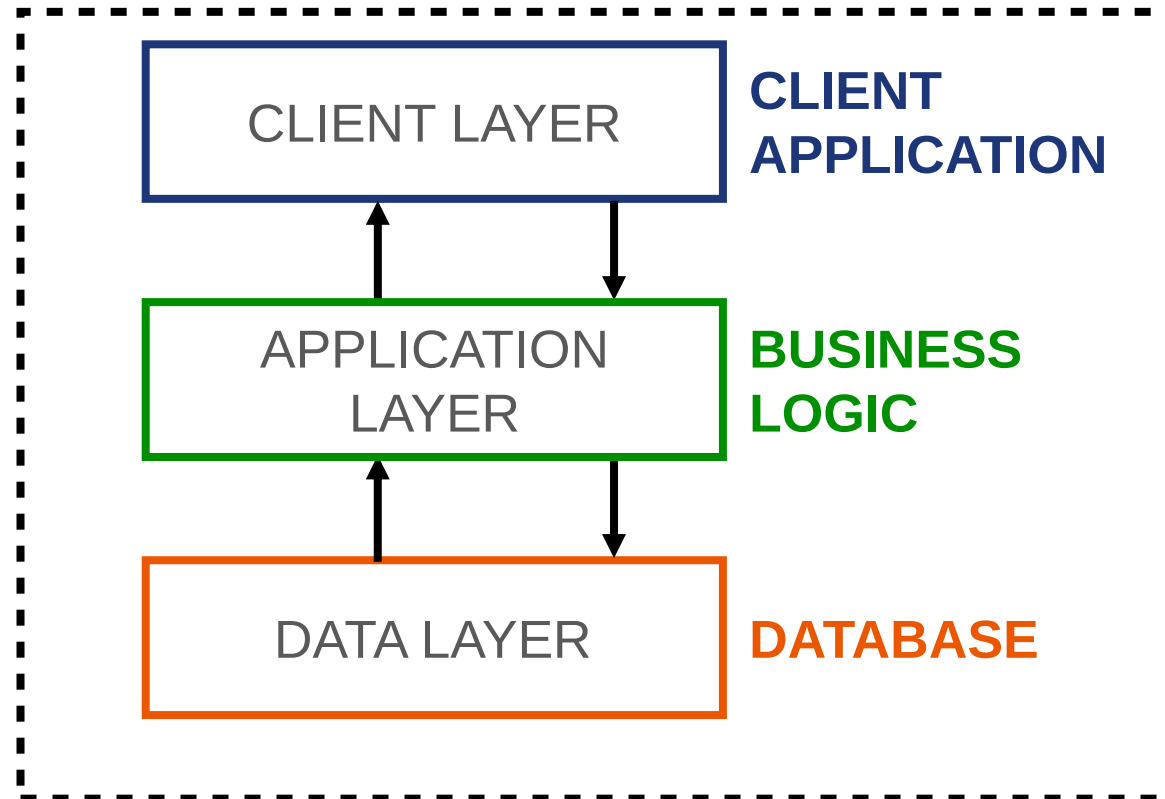
SAP Netweaver

Is the base for any other application of SAP, as the ERP. It enables the development and integration of different applications. Modern versions support also IoT and AI for predictive analysis.

SAP NETWEAVER WEB APPLICATION SERVER

The three layers
may run on the
same or different
machines

Three-Tier Architecture



SAP ABAP

ABAP programs (*Advanced Business Application Programming*) are programs written in a specific language developed by SAP and used to develop applications on SAP systems.

ABAP is used to customize and create specific solutions inside of the SAP ERP system.

ABAP programs run within the SAP ERP system, i.e., the **SAP NetWeaver Web Application Server**:

- ABAP is strictly integrated with the SAP central database (where **also programs** reside).
- Several **applications** (instances) connected to the database.
- **Database Interface** translating ABAP “Open SQL” (a SAP dialect for SQL) statements into “Native SQL” statements

SAP ABAP: Open SQL

An example of Open SQL:

```
SELECT num1 + num2 AS sum, COUNT( * ) AS  
count  
  
FROM demo_expressions  
GROUP BY num1, num2  
ORDER BY sum  
INTO TABLE @DATA(result1)
```

SAP ABAP: a Program

An example of ABAP program:

```
report my_report.

tables: table_1, table_2.

select * from table_1 where field_1 eq 'STRING' and
      field_2 eq 'TABL' and field_3 like 'M%'.

select * from table_2 where field_1 eq 'STRING'.
      write: / table_1-field_1, table_2-field_2.
      exit.
endselect.

endselect.
```

Main Modules:

- Financial Accounting (FI)
- Controlling (CO)
- Asset Accounting (AA)
- Sales & Distribution (SD)
- Material Management (MM)
- Product Planning (PP)
- Quality Management (QM)
- Project System (PS)
- Plant Maintenance (PM)
- Human Resources (HR)

SAP Timeline



Main Releases

- SAP R/1 System RF — 1973
- SAP R/2 Mainframe System — 1979
- SAP R/3 Enterprise Edition 1.0 A — 1992
- ...
- SAP R/3 ENTERPRISE 47X200 — 2003
- SAP ERP Central Component (ECC) 5.0 — 2004
- SAP ERP Central Component (ECC) 6.0 — 2006

Small enterprises

- SAP Business One

Medium enterprises

- SAP Business ByDesign
- SAP S/4HANA

Large enterprises

- SAP S/4HANA (Cloud)
- SAP ERP

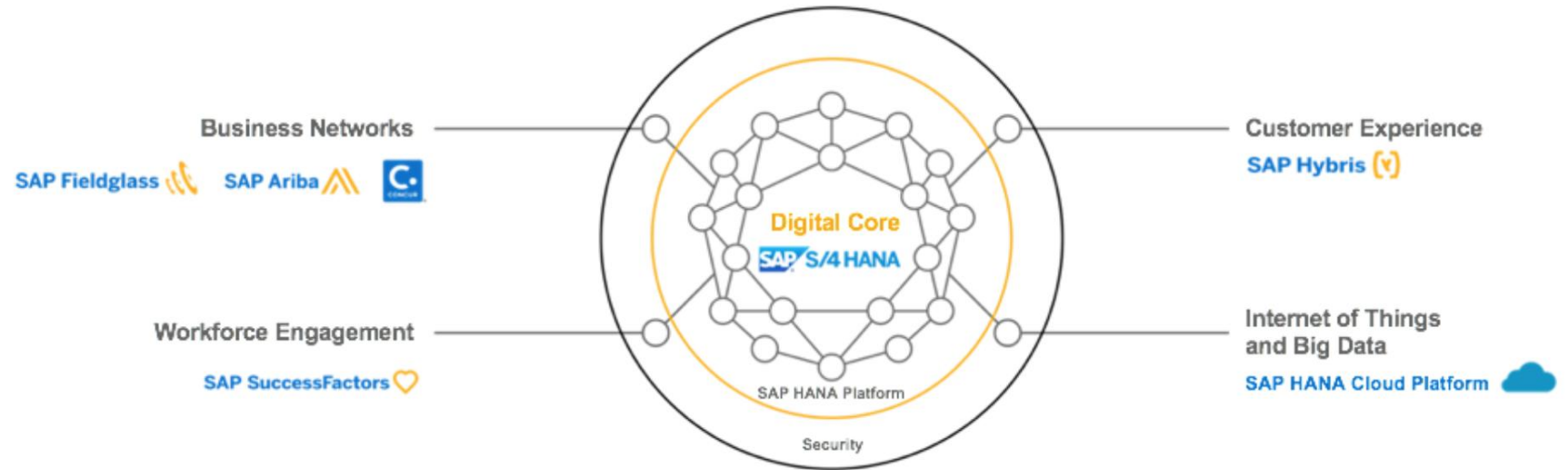
SAP S/4HANA

SAP S/4HANA — Tries to integrate...

- Business networks
- The Internet of Things
- Big Data

Towards **intelligent** ERP systems!

Towards cloud and mobile solutions.



Immediate

Empowering business users with insights to act in the moment.



Intelligent

Beyond automation, to predictive suggestion.



Integrated

Not only between your departments, but connected to the world.

SAP S/4HANA



SAP HANA is the **database** part of the ERP system:

- An **in-memory** database (cost?)
- A **column-oriented** database
- **Scalable** to cloud systems
- Combining OLAP and OLTP operations (**OLTAP...**)
- Integrating **predictive** analytics (machine learning!)
- Integrating **R and Python procedures!**

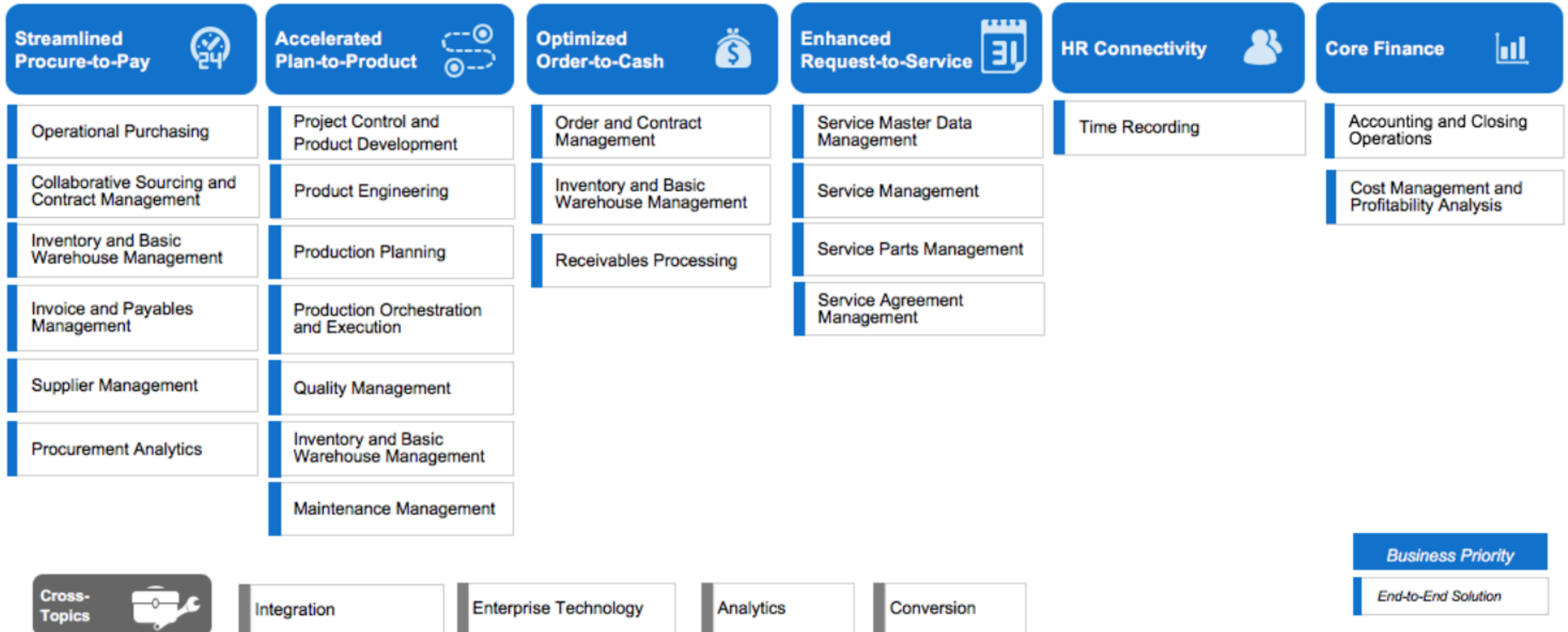
Features:

- **Active** and **passive** storage
- Enabling massive **data compression**
- Enabling advanced **parallelism**
- **Data virtualization**, i.e., the possibility to access data from different sources without physically copy or move them (vs. ETL)
- **Aggregation on-the-fly**
- SQL Script for **complex queries**
- Manage both **structured** and **less structured** data

SAP S/4HANA

SAP S/4HANA 1610

Value map for the digital core*



Free 14-day trial available at

<https://www.sap.com/cmp/oth/crm-s4hana/s4hana-on-premise.html>